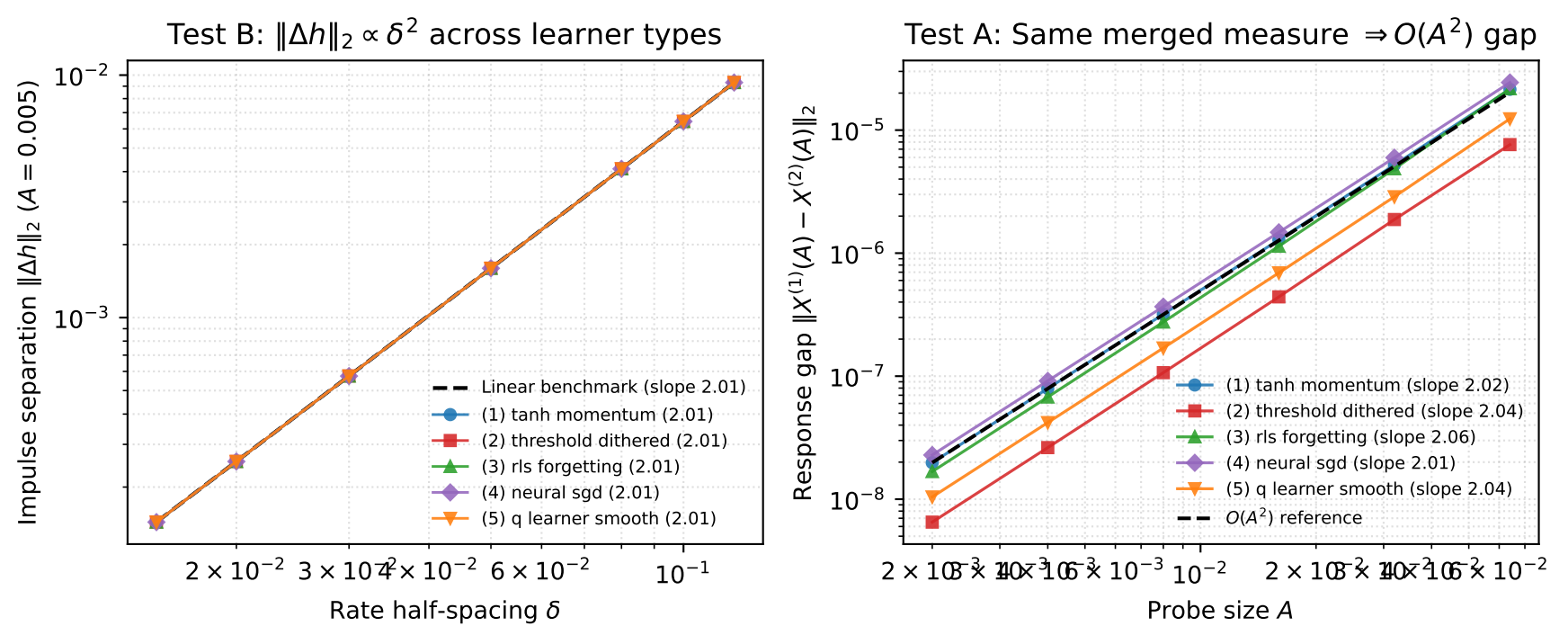
Test B: $\|\Delta h\|_2 \propto \delta^2$ across learner types
Impulse separation $\|\Delta h\|_2$ ($A = 0.005$)
10^{-2}
10^{-3}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
10^{-1}
Rate half-spacing δ
Linear benchmark (slope 2.01)
(1) tanh momentum (2.01)
(2) threshold dithered (2.01)
(3) rls forgetting (2.01)
(4) neural sgd (2.01)
(5) q learner smooth (2.01)
Test A: Same merged measure $\Rightarrow O(A^2)$ gap
Response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$
10^{-5}
10^{-6}
10^{-7}
10^{-8}
2×10^{-3}
3×10^{-3}
4×10^{-3}
6×10^{-3}
10^{-2}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
Probe size A
(1) tanh momentum (slope 2.02)
(2) threshold dithered (slope 2.04)
(3) rls forgetting (slope 2.06)
(4) neural sgd (slope 2.01)
(5) q learner smooth (slope 2.04)
$O(A^2)$ reference